

模板标题

作者

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Problem 1

代码:

```
1 def permutations(a, arr, pos, end): # 获取列表的全排列
2     if pos == end: # 递归结束
3         arr = arr[:]
4         a.append(arr)
5     else:
6         for index in range(pos, end):
7             arr[index], arr[pos] = arr[pos], arr[index] # 交↔
              换相邻两个元素
8             permutations(a, arr, pos+1, end) # 列表长度-1
9             arr[index], arr[pos] = arr[pos], arr[index] # 交↔
              换相邻两个元素
10    return a
```

Problem 2

代码:

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 from matplotlib import cm
4 import math
5 from matplotlib.pylab import mpl
6 from mpl_toolkits.mplot3d import Axes3D
7
8 u = np.linspace(-3 , 3 , 101)
```

```
9 x , y = np.meshgrid(u , u)
10 print("x = \n" , x)
11 print("y = \n" , y)
12 z = np.exp(-(x**2 + y**2) / 2) / math.sqrt(2 * math.pi)
13
14 fig = plt.figure()
15 ax = fig.add_subplot(111, projection = '3d')
16 ax.plot_surface(x , y , z, rstride = 2, cstride = 2 , cmap = ←
    cm.get_cmap('winter'), linewidth = 0.5)
17 plt.show()
```

输出图片如下：

